

1. Hadoop Setup Using Cloudera

What is Hadoop?

Apache Hadoop is an open-source framework used for distributed storage and processing of large datasets.

Main Components:

- HDFS → Storage
 - MapReduce → Processing
 - YARN → Resource management
-

Steps to Setup Hadoop Using Cloudera VM

Step 1: Download Cloudera QuickStart VM

Use:

- VirtualBox or VMware
- Cloudera QuickStart VM

Official website:

[Cloudera Official Website](#)

Step 2: Start Virtual Machine

- Open VirtualBox
- Import Cloudera VM
- Start VM

Default credentials:

Username: cloudera

Password: cloudera

Step 3: Start Hadoop Services

Open terminal and run:

```
sudo service hadoop-hdfs-namenode start  
sudo service hadoop-hdfs-datanode start
```

```
sudo service hadoop-yarn-resourcemanager start  
sudo service hadoop-yarn-nodemanager start
```

Step 4: Check Hadoop Version

```
hadoop version
```

Step 5: Create Directory in HDFS

```
hdfs dfs -mkdir /input
```

Step 6: Upload File to HDFS

```
hdfs dfs -put sample.txt /input
```

Step 7: View Files

```
hdfs dfs -ls /input
```

Viva Questions for Hadoop

Q1. What is HDFS?

Hadoop Distributed File System used for distributed storage.

Q2. What is NameNode?

Master node managing metadata.

Q3. What is DataNode?

Stores actual data blocks.

Q4. What is YARN?

Resource manager in Hadoop.

2. Google Cloud BigQuery

What is BigQuery?

Google BigQuery is Google's serverless cloud data warehouse used for big data analytics.

Steps for BigQuery Setup

Step 1: Open Google Cloud Platform

[Google Cloud Platform](#)

Step 2: Create Project

- Click “New Project”
 - Enter project name
-

Step 3: Enable BigQuery API

- Search “BigQuery”
 - Enable API
-

Step 4: Open BigQuery Console

- Create Dataset
 - Create Table
-

Step 5: Run SQL Query

Example:

```
SELECT name, age  
FROM employees  
WHERE age > 25;
```

Viva Questions for BigQuery

Q1. Is BigQuery serverless?

Yes.

Q2. Which language is used in BigQuery?

SQL.

Q3. Main advantage?

Scalable analytics with no infrastructure management.

3. Databricks Lakehouse Platform

What is Databricks?

Databricks is a cloud-based data engineering and machine learning platform based on Apache Spark.

Setup Steps

Step 1: Open Databricks Community Edition

[Databricks Community Edition](#)

Step 2: Create Workspace

- Sign in
 - Create cluster
-

Step 3: Create Notebook

Languages supported:

- Python
 - SQL
 - Scala
 - R
-

Step 4: Sample PySpark Code

```
data = [("Ayesha", 21), ("Shraddha", 22)]
```

```
df = spark.createDataFrame(data, ["Name", "Age"])
```

```
df.show()
```

Viva Questions for Databricks

Q1. What is Spark?

Distributed processing engine.

Q2. What is Lakehouse?

Combination of Data Lake + Data Warehouse.

Q3. Which engine powers Databricks?

Apache Spark.

4. Snowflake

What is Snowflake?

Snowflake is a cloud-based data warehousing platform.

Setup Steps

Step 1: Create Account

[Snowflake Official Website](#)

Step 2: Create Database

CREATE DATABASE company;

Step 3: Create Table

```
CREATE TABLE employee (  
  id INT,  
  name STRING,  
  salary INT  
);
```

Step 4: Insert Values

```
INSERT INTO employee  
VALUES (1, 'Ayesha', 50000);
```

Step 5: Retrieve Data

```
SELECT * FROM employee;
```

Viva Questions for Snowflake

Q1. Is Snowflake cloud-native?

Yes.

Q2. What language does Snowflake use?

SQL.

Q3. Advantage of Snowflake?

Automatic scaling and separation of storage/computation.

5. Amazon Redshift

What is Redshift?

Amazon Redshift is AWS's cloud data warehouse service.

Setup Steps

Step 1: Open AWS Console

[Amazon Web Services](#)

Step 2: Open Redshift Service

- Create Cluster
 - Configure database
-

Step 3: Connect Using Query Editor

Step 4: Create Table

```
CREATE TABLE student(  
  id INT,  
  name VARCHAR(50)  
);
```

Step 5: Insert Data

```
INSERT INTO student VALUES(1, 'Ayesha');
```

Step 6: Retrieve Data

```
SELECT * FROM student;
```

Viva Questions for Redshift

Q1. Which company developed Redshift?

Amazon AWS.

Q2. What is Redshift used for?

Large-scale data analytics.

Q3. Is Redshift relational?

Yes.

Short Conclusion for Practical File

Thus, Hadoop and cloud-based big data platforms like BigQuery, Databricks, Snowflake, and Redshift were successfully configured and basic operations were performed.

1. Hadoop Installation Commands (Ubuntu/Linux)

Update System

```
sudo apt update  
sudo apt upgrade -y
```

Install Java

```
sudo apt install openjdk-11-jdk -y
```

Check Java version:

```
java -version
```

Create Hadoop User

`sudo adduser hadoop`

Switch user:

`su - hadoop`

Install SSH

`sudo apt install ssh -y`

Generate SSH Key:

`ssh-keygen -t rsa -P ""`

Enable SSH:

`cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`

`chmod 600 ~/.ssh/authorized_keys`

Test SSH:

`ssh localhost`

Download Hadoop

[Apache Hadoop Downloads](https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz)

Command:

`wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz`

Extract:

`tar -xvzf hadoop-3.3.6.tar.gz`

Move Hadoop:

`sudo mv hadoop-3.3.6 /usr/local/hadoop`

Configure Environment Variables

Open bashrc:

`nano ~/.bashrc`

Add:

```
export HADOOP_HOME=/usr/local/hadoop
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Apply changes:

```
source ~/.bashrc
```

Verify Hadoop

```
hadoop version
```

Format NameNode

```
hdfs namenode -format
```

Start Hadoop Services

```
start-dfs.sh
start-yarn.sh
```

Check Running Processes

```
jps
```

You should see:

- NameNode
 - DataNode
 - ResourceManager
 - NodeManager
-

2. Cloudera Installation Commands

Install VirtualBox

```
sudo apt install virtualbox -y
```

Download Cloudera VM

[Cloudera Downloads](#)

Import VM

Usually done graphically in VirtualBox:

- File → Import Appliance
-

Start Hadoop Services in Cloudera

```
sudo service hadoop-hdfs-namenode start
sudo service hadoop-hdfs-datanode start
sudo service hadoop-yarn-resourcemanager start
sudo service hadoop-yarn-nodemanager start
```

3. Google Cloud BigQuery Commands

Install Google Cloud SDK

[Google Cloud SDK Installation Guide](#)

Ubuntu commands:

```
sudo apt install apt-transport-https ca-certificates gnupg curl -y
curl https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo apt-key add -
echo "deb https://packages.cloud.google.com/apt cloud-sdk main" | sudo tee -a
/etc/apt/sources.list.d/google-cloud-sdk.list
sudo apt update && sudo apt install google-cloud-sdk -y
```

Initialize Google Cloud

```
gcloud init
```

Authenticate

```
gcloud auth login
```

Install BigQuery CLI

```
sudo apt install bq -y
```

Create Dataset

```
bq mk mydataset
```

Run Query

```
bq query --use_legacy_sql=false 'SELECT "Hello BigQuery"'
```

4. Databricks Installation Commands

Install Python

```
sudo apt install python3 python3-pip -y
```

Install Java

```
sudo apt install openjdk-11-jdk -y
```

Install Apache Spark

```
wget https://downloads.apache.org/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz
```

Extract:

```
tar -xvzf spark-3.5.1-bin-hadoop3.tgz
```

Move:

```
sudo mv spark-3.5.1-bin-hadoop3 /opt/spark
```

Configure Spark

```
nano ~/.bashrc
```

Add:

```
export SPARK_HOME=/opt/spark
export PATH=$PATH:$SPARK_HOME/bin
```

Apply:

```
source ~/.bashrc
```

Verify Spark

```
spark-shell
```

Install Databricks CLI

```
pip install databricks-cli
```

Configure Databricks

```
databricks configure --token
```

5. Snowflake Installation Commands

Install Python Pip

```
sudo apt install python3-pip -y
```

Install Snowflake Connector

```
pip install snowflake-connector-python
```

Verify Installation

```
python3
```

Inside Python:

```
import snowflake.connector
```

If no error appears, installation is successful.

6. Amazon Redshift Commands

Install PostgreSQL Client

PostgreSQL is used to connect with Redshift.

```
sudo apt install postgresql-client -y
```

Connect to Redshift Cluster

```
psql -h your-redshift-endpoint -U username -d database
```

Example:

```
psql -h redshift-cluster.amazonaws.com -U admin -d dev
```

Run SQL Commands

Create table:

```
CREATE TABLE employee(  
  id INT,  
  name VARCHAR(50)  
);
```

Insert:

```
INSERT INTO employee VALUES(1, 'Ayesha');
```

Retrieve:

```
SELECT * FROM employee;
```

Important Viva Questions

Q1. Why is Java required for Hadoop?

Because Hadoop is Java-based.

Q2. What does jps command do?

Shows running Java processes.

Q3. Which command starts Hadoop DFS?

start-dfs.sh

Q4. Which command starts YARN?

start-yarn.sh

Q5. Which CLI tool is used for Google Cloud?

gcloud

Q6. Which language is mainly used in Spark?

Scala/Python.

Q7. Which protocol is used for Redshift connection?

PostgreSQL protocol.

